P_2 AUX 17

P2. **[Solenoides]**

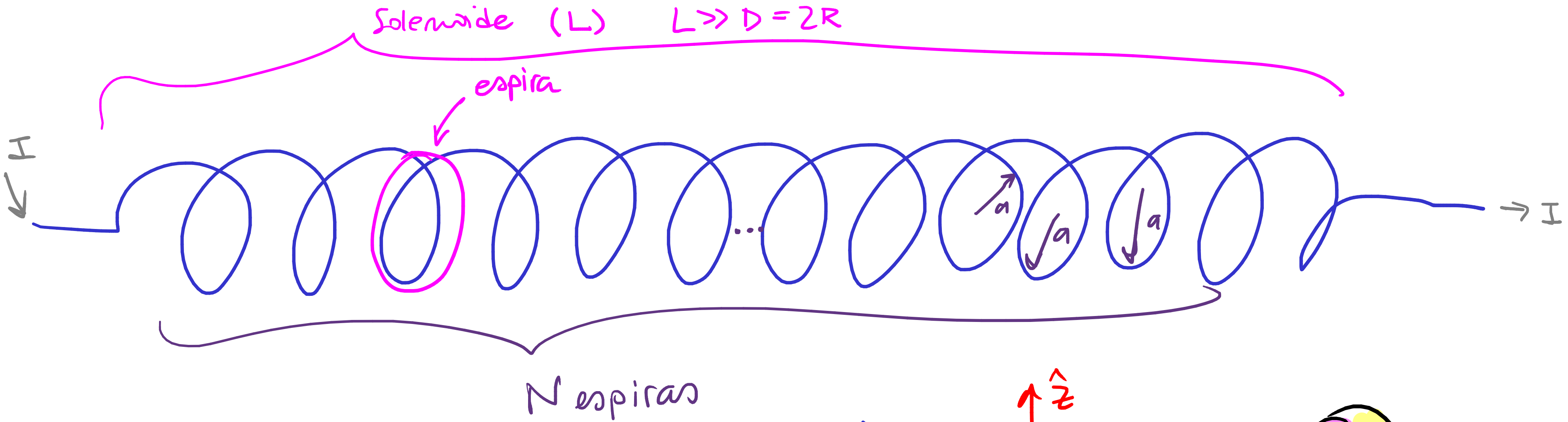
Se tiene un alambre lo suficientemente largo tal que al enrollarlo en N espiras radio a, el largo del solenoide generado es mucho mayor a su diámetro. Si transporta una corriente neta I, calcule el campo magnético en todo el espacio cuando:

(a) Se tiene un caso ideal.

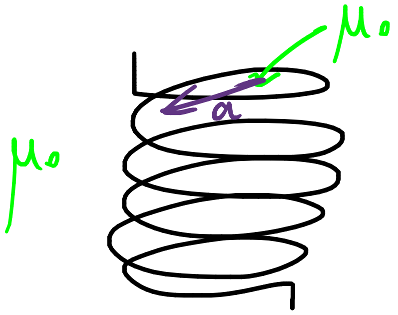
(b) Al interior, se distribuyen por igual (angularmente) dos medios de permeabilidades μ_1 y μ_2.

(i) $0<r<a$

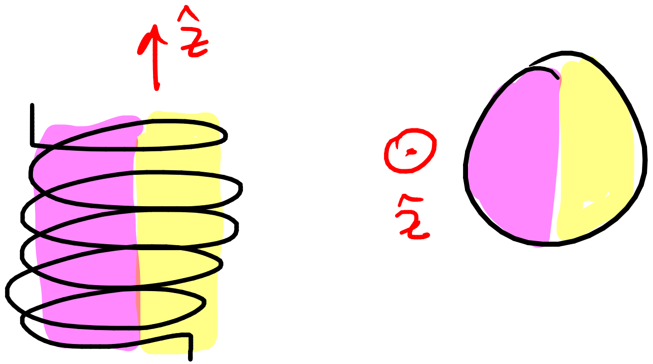
(ii) $a>r$

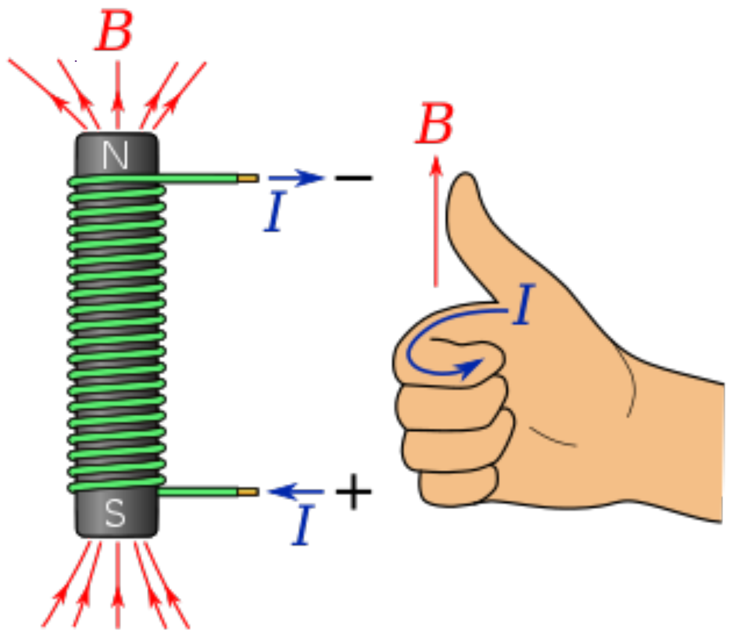


a)



b)





B
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I
−
B
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S
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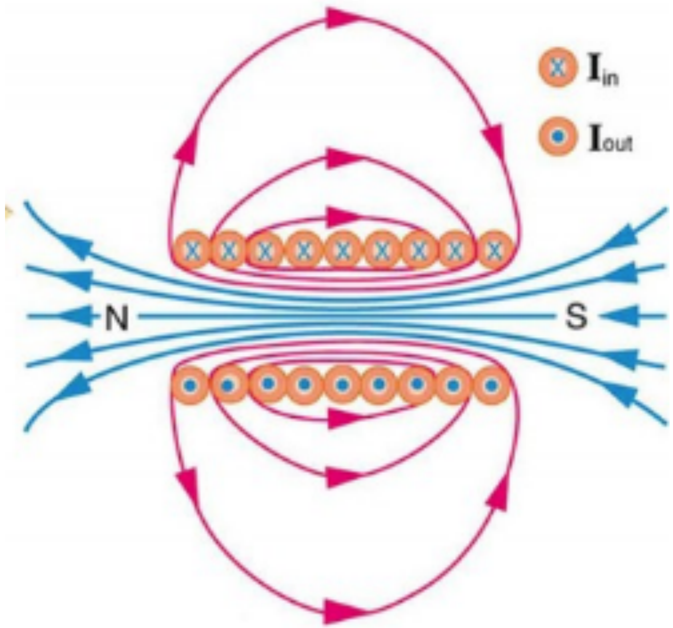
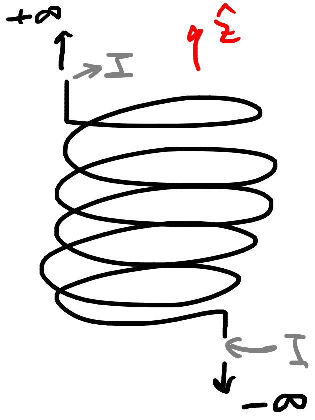
+

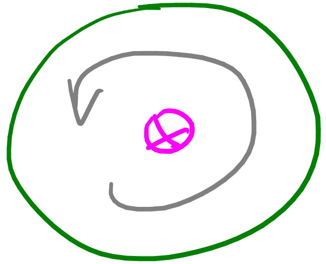


$\mathbf{I}_{in}$
$\mathbf{I}_{out}$
N
S

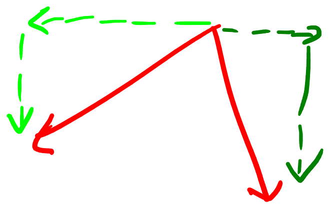
P2) a)



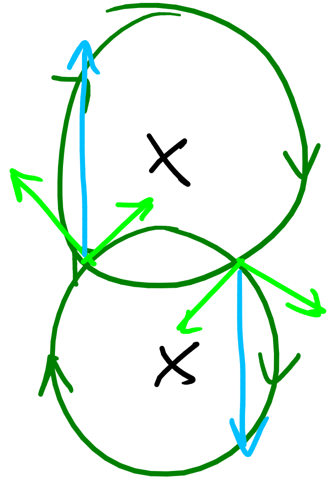
$\odot\ \hat{z}$

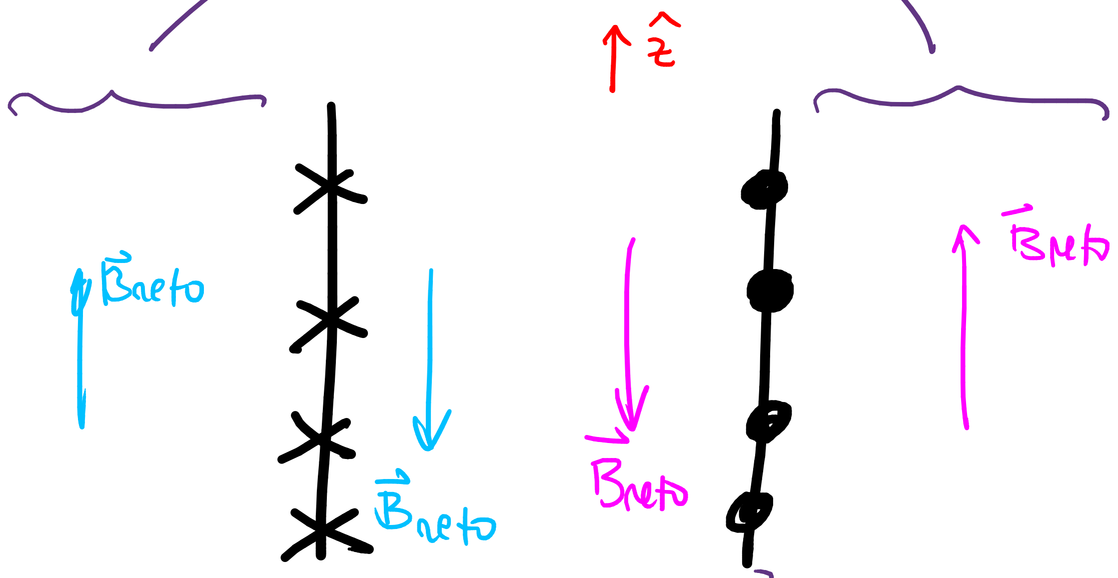


$$\vec{B} = B(r,\cancel{\theta},\cancel{z})(-\hat{z})$$

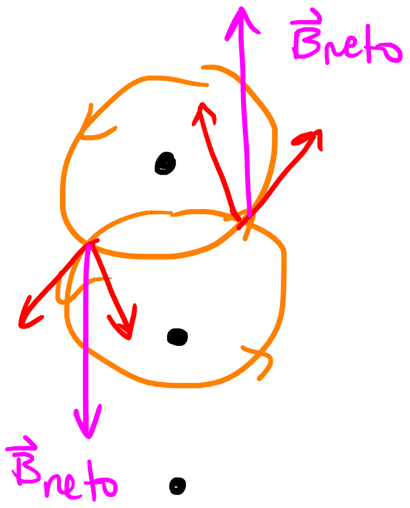


$$\vec{B} \approx \vec{0}$$





se intensifica



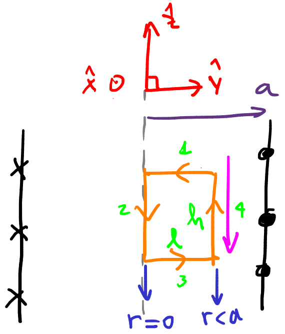
i) $0<r<a$

Ley de Ampère: $\oint_\Gamma \vec{B}\cdot d\vec{l} = \mu_0 I_{enlazada}$

0°) justificar

1°) elegir loop

* coherente con $\vec{B}$
* enlazar corriente



$\Gamma_1: d\vec{l} = dy\,\hat{y}$; $y: l\to 0$

$\Gamma_2: d\vec{l} = dz\,\hat{z}$; $z: h\to 0$ $\quad r=0$

$\Gamma_3: d\vec{l} = dy\,\hat{y}$; $y: 0\to l$

$\Gamma_4: d\vec{l} = dz\,\hat{z}$; $z: 0\to h$ $\quad r<a$

$\Gamma = \Gamma_1 \cup \Gamma_2 \cup \Gamma_3 \cup \Gamma_4$

L.I.

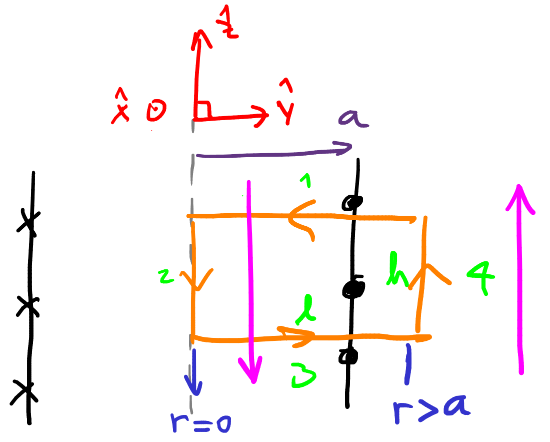
$$\oint_\Gamma \vec{B}\cdot d\vec{l} = \cancel{\int_l^0 B(r)\underbrace{(-\hat{z})\cdot dy\,\hat{y}}_{0}} + \int_h^0 B(r=0)\underbrace{(-\hat{z})\cdot dz\,\hat{z}}_{-1} + \cancel{\int B(r)\underbrace{(-\hat{z})\cdot dy\,\hat{y}}_{0}} + \int_0^h B(r<a)\underbrace{(-\hat{z})\cdot dz\,\hat{z}}_{-1}$$

$$= -B(r=0)\int_h^0 dz - B(r<a)\int_0^h dz = -B(r=0)[-h] - B(r<a)[h] = B(r=0)h - B(r<a)h$$

L.D.

$I_{enlazada} = 0$

$$\Rightarrow \big(B(r=0) - B(r<a)\big)h = 0 \Rightarrow \boxed{B(r=0) = B(r<a)} \quad (1)$$



$$\frac{N}{L} = \frac{x}{h} \Leftrightarrow x = \frac{N}{L}\, h \qquad \frac{N}{L} =: n = \frac{\#\text{vueltas}}{\text{largo}}$$

L.I

$$\oint_{\Gamma} \vec{B}\cdot d\vec{l} = \cancelto{0}{\int_{\Gamma_1} \vec{B}\cdot d\vec{l}}_{\vec{B}\perp d\vec{l}} + \int_h^0 B(r=0)\underbrace{(-\hat{z})\cdot dz\,\hat{z}}_{-1} + \cancelto{0}{\int_{\Gamma_3} \vec{B}\cdot d\vec{l}}_{\vec{B}\perp d\vec{l}} + \cancelto{0}{\int_0^h \underbrace{B(r>a)}_{\approx 0}\,\hat{z}\cdot dz\,\hat{z}}$$

$$= -B(r=0)\int_h^0 dz = -B(r=0)[-h] = B(r=0)\,h$$

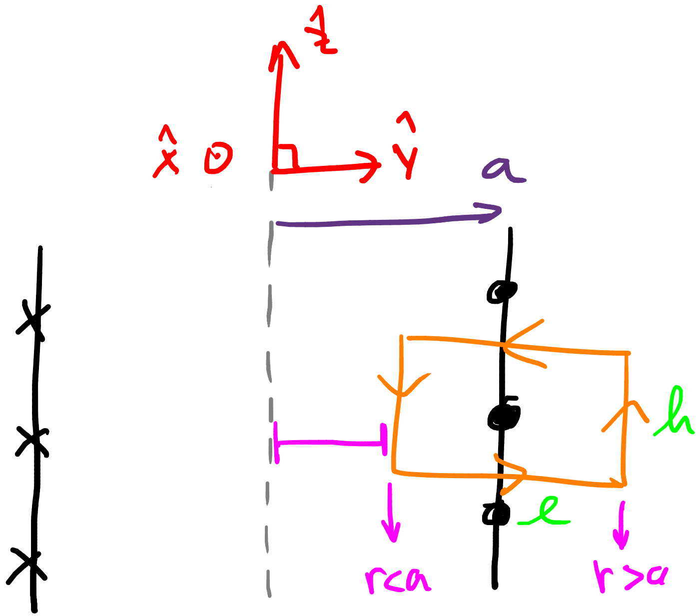
L.D.

$$I_{\text{enlazada}} = I\,\frac{N}{L}\,h = I\,n\,h$$

— o —

$$\Rightarrow B(r=0)\,h = \mu_0 I n h \Rightarrow \boxed{B(r=0) = \mu_0 I n} \quad (2)$$

$$\vec{B}(r=0) = \boxed{\vec{B}(r<a) = \mu_0 I n\,(-\hat{z})}$$



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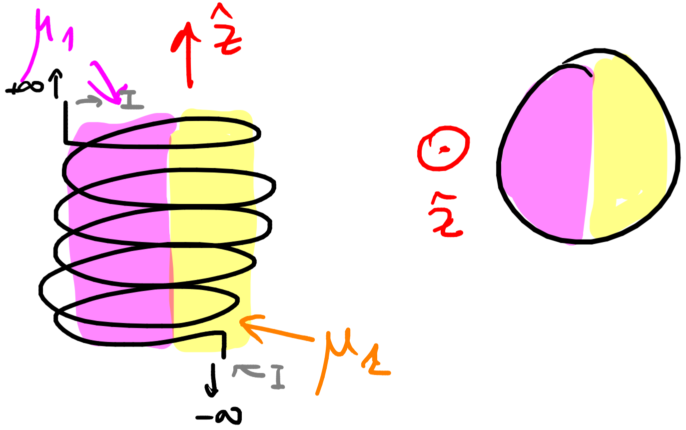
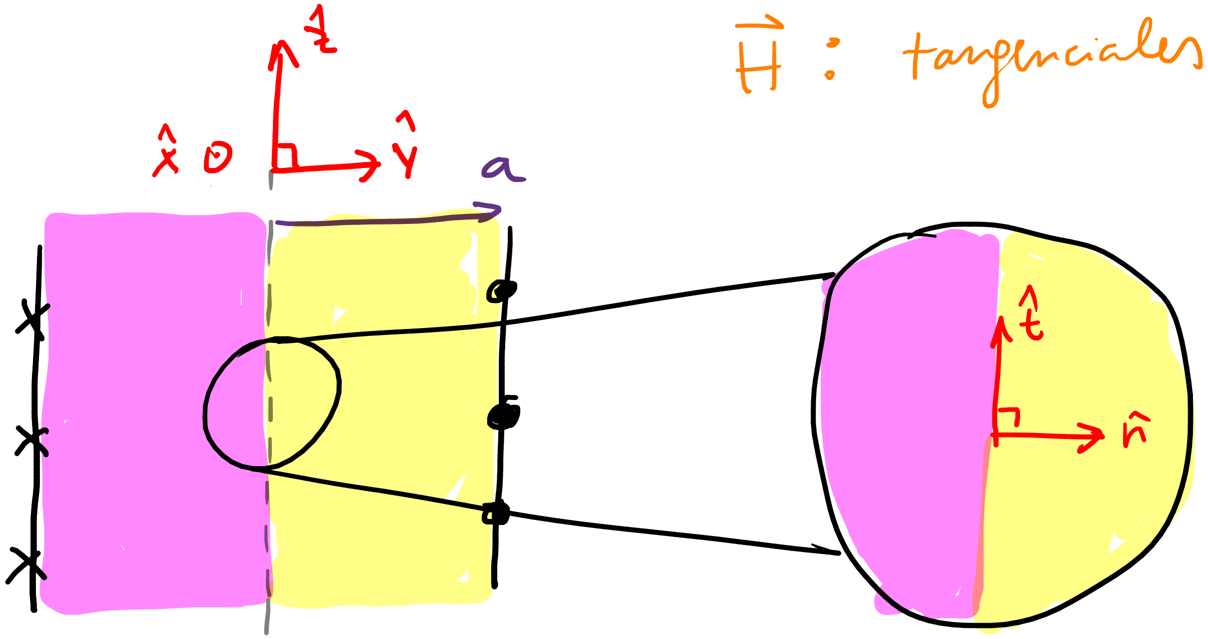
r>a

(P2) b)



Ley de Ampère: $\oint_\Gamma \vec{B}\cdot d\vec{l} = \mu_0 \underbrace{I_{enlazada}}_{(I_{libre} + I_m)}$

Ley de Ampère generalizada: $\oint_\Gamma \vec{H}\cdot d\vec{l} = I_{enlazada\ libre}$



$\vec{H}$: tangenciales $\quad H_{2t} - H_{1t} = \| \cancel{\vec{K} \times \hat{n}} \| \rightarrow 0$

$\Leftrightarrow H_{2t} = H_{1t}$

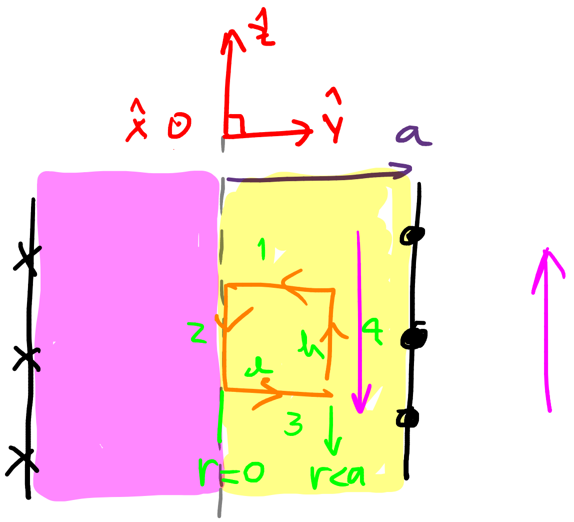
$\hat{t} := \hat{z}$

$\hat{n} := \hat{y}$ (local)

($:= \hat{r}$ en general)

$H_{1t} = H_{2t}$

$\Leftrightarrow H_{1z} = H_{2z} = H$



L.I.)

$$\oint_{\Gamma} \vec H\cdot d\vec l = \underbrace{\cancel{\int_{\Gamma_1} \vec H \cdot d\vec l}}_{\vec H \perp d\vec l \;\Rightarrow\; 0} + \int_{h}^{0} H(r=0)(-\hat z)\cdot dz\,\hat z + \underbrace{\cancel{\int_{\Gamma_3} \vec H\cdot d\vec l}}_{\vec H \perp d\vec l \;\Rightarrow\; 0} + \int_{0}^{h} H(r<a)(-\hat z)\cdot dz\,\hat z$$

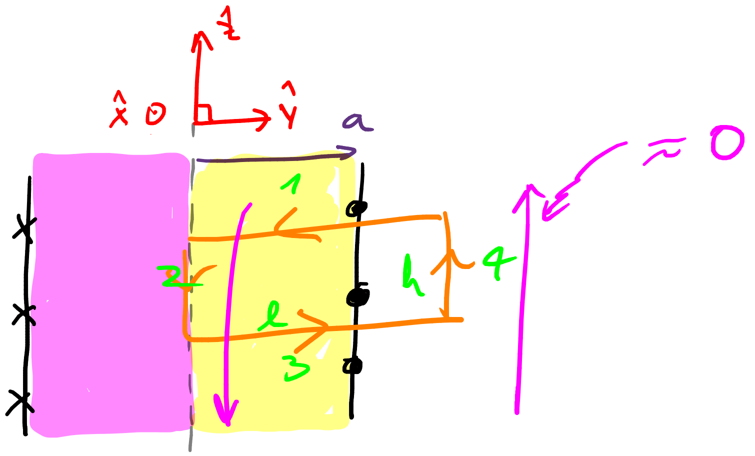
$$= -H(r=0)[-h] - H(r<a)[h] = \big(H(r=0)-H(r<a)\big)h$$

L.D.)

$I_{enlazada} = 0$

— o —

$$\Rightarrow \big(H(r=0)-H(r<a)\big)h = 0 \Rightarrow \boxed{H(r=0) = H(r<a)} \quad (1)$$



L.I.)

$$\oint_{\Gamma} \vec{H}\cdot d\vec{l} = -H(r=0)[-h] = H(r=0)\,h$$

L.D.)

$$I_{\text{enlazada}} = I\frac{N}{L}h = Inh$$

-o-

$$\Rightarrow H(r=0)\cancel{h} = In\cancel{h} \Rightarrow \boxed{H(r=0) = In} \quad (2)$$

$$(2) \Leftrightarrow (1) \quad \vec{H}(r<a) = In(-\hat{z})$$

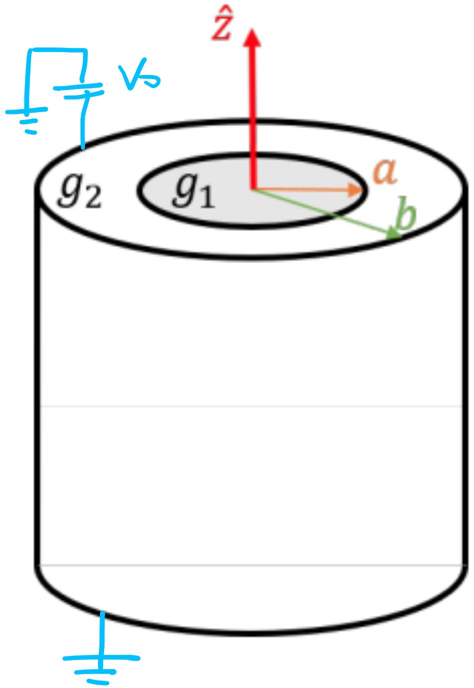
$$\vec{B} = \mu\vec{H} \Rightarrow \vec{B}_1 = \mu_1 In(-\hat{z})$$

$$\vec{B}_2 = \mu_2 In(-\hat{z})$$

P_3 AUX 17

P3. [Medios Magnéticos]

Un cable rectilíneo infinito está formado por dos materiales conductores coaxiales: el núcleo, de radio a y conductividad g_1, y el recubrimiento, de radio b y resistividad $1/g_2$. Considere un trozo de largo L, cuyo extremo superior está a potencial V_0 y el inferior está conectado a tierra. Determine:



(a) La densidad de corriente $\vec{J}$ en todo el alambre.

(b) Los campos intensidad magnética $\vec{H}$ y magnetización $\vec{M}$ en todo el espacio, sabiendo que el cable tiene susceptibilidad magnética χ. ← chi

(c) La densidad volumétrica de corriente de magnetización $\vec{J}_m$.

(d) La(s) densidad(es) superficial(es) de corriente de magnetización $\vec{K}_m$.

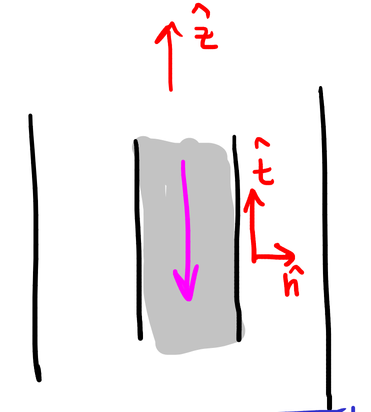
$\vec{J} = J(r)(-\hat{z})$

$\cancel{\theta}, \cancel{z}$

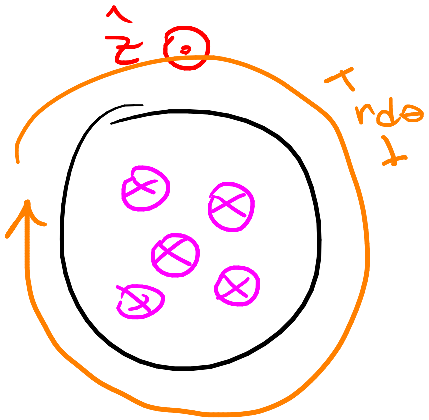
(i) $0<r<a$

(ii) $a<r<b$

(iii) $r>b$



$\boxed{J_1 g_1 = J_2 g_2}$



P_3 a)

$$\nabla \cdot \vec{J} + \cancelto{0}{\frac{\partial \rho}{\partial t}} = 0$$

$$\Rightarrow \nabla \cdot \vec{J} = 0 \quad ; \quad \vec{J} = J(r)(-\hat{z})$$

$$\Rightarrow \frac{\partial J_z}{\partial z} = 0$$

$$\Rightarrow J_z = J_0 \,, \quad J_0 \in \mathbb{R}$$

$$\Rightarrow \vec{J} = J_0(-\hat{z})$$

$$\therefore \vec{J} = \frac{V_0 g}{L}(-\hat{z})$$

$$\boxed{\vec{J}_1 = \frac{V_0 g_1}{L}(-\hat{z})} \qquad \boxed{\vec{J}_2 = \frac{V_0 g_2}{L}(-\hat{z})}$$

$0 < r < a$ $\qquad$ $a < r < b$

d.d.p.: $\underbrace{V(a) - V(b)}_{=:\Delta V} = -\int_b^a \vec{E} \cdot d\vec{l}$

$$\cancelto{V_0}{V(L)} - \cancelto{0}{V(0)} = -\int_0^L E(r)\hat{z} \cdot dz\hat{z}$$

$$\Leftrightarrow V_0 = -E(r)\int_0^L dz \quad ; \quad \vec{J} = g\vec{E}$$

$$\Leftrightarrow V_0 = -\left(\frac{1}{g}(-J_0)\right)L \qquad \Leftrightarrow \vec{E} = \frac{1}{g}\vec{J}$$

$$\Rightarrow E = \frac{1}{g}(-J_0)$$

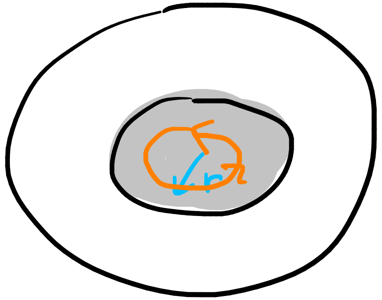
$$\Leftrightarrow V_0 = \frac{1}{g} J_0 L$$

$$\Leftrightarrow J_0 = \frac{V_0 g}{L}$$

$$\boxed{J_1 g_1 = J_2 g_2}$$

$$\frac{J_1}{J_2} = \frac{g_1}{g_2} \Leftrightarrow \frac{V_0 g_1}{L} \cdot \frac{L}{V_0 g_2}$$

P3 b) Ley de Ampère generalizada: $\oint_{\Gamma} \vec{H}\cdot d\vec{l} = I_{\substack{\text{enlazada}\\ \text{libre}}}$



(i) $0<r<a$

L.I

$$\oint_{\Gamma} \vec{H}\cdot d\vec{l} = \int_0^{2\pi} H(r)\hat{\theta}\cdot r\,d\theta\,\hat{\theta} = H(r)\,r\int_0^{2\pi} d\theta = H(r)\cdot 2\pi r$$

L.D.

$$I_{\substack{\text{enlazada}\\ \text{libre}}} = \iint \vec{J}_1\cdot d\vec{S} = \int_0^{2\pi}\int_0^{r\,(<a)} \|J_1\|\,\hat{z}\cdot r\,dr\,d\theta\,\hat{z} = \|J_1\|\cdot\frac{1}{2}r^2\cdot 2\pi = \pi\|J_1\|r^2$$

— o —

$$\Rightarrow H(r)\cdot 2\pi r = \pi\|J_1\|r^2$$

$$\Rightarrow \boxed{\vec{H} = \frac{1}{2}\|J_1\|\,r\,\hat{\theta}}\ ,\quad 0<r<a$$

$$\vec{M} = \left(\frac{\mu}{\mu_0}-1\right)\vec{H}$$

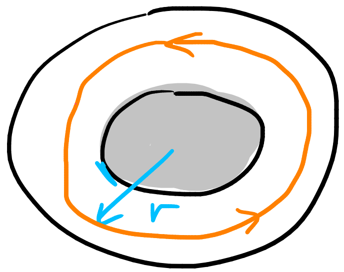
$$\mu = \mu_0 K_m$$

$$\chi = K_m - 1$$

$$\Leftrightarrow \vec{M} = \chi\vec{H}$$

$$\boxed{\vec{M} = \frac{1}{2}\chi\|J_1\|\,r\,\hat{\theta}}\ ,\quad 0<r<a$$

(ii) $a<r<b$



L.I

$$\oint_{\Gamma} \vec{H}\cdot d\vec{l} = \int_0^{2\pi} H(r)\hat{\theta}\cdot r\,d\theta\,\hat{\theta} = H(r)\,r\int_0^{2\pi} d\theta = H(r)\cdot 2\pi r$$

L.D.

$$I_{\text{enlazada libre}} = \int_0^{2\pi}\int_0^{a} \|J_1\|\hat{z}\cdot r\,dr\,d\theta\,\hat{z} + \int_0^{2\pi}\int_a^{r} \|J_2\|\hat{z}\cdot r\,dr\,d\theta\,\hat{z}$$

(a<r<b)

$$= \pi\|J_1\|a^2 + \|J_2\|\cdot\cancel{2}\pi\cdot\frac{1}{\cancel{2}}(r^2-a^2)$$

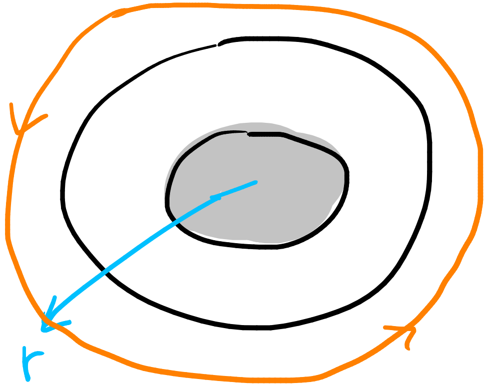
— o —

$$\Rightarrow H(r)\cdot 2\pi r = \pi\|J_1\|a^2 + \|J_2\|\pi(r^2-a^2)$$

$$\Rightarrow \boxed{\vec{H} = \frac{\|J_1\|a^2}{2}\frac{1}{r} + \frac{\|J_2\|}{2}\frac{r^2-a^2}{r}\,\hat{\theta}} \quad (a<r<b)$$

$$\boxed{\vec{M} = \chi\left(\frac{\|J_1\|a^2}{2}\frac{1}{r} + \frac{\|J_2\|}{2}\frac{r^2-a^2}{r}\right)\hat{\theta}}$$

(iii) $r>b$



L.I

$$\oint_\Gamma \vec{H}\cdot d\vec{l} = \int_0^{2\pi} H(r)\hat{\theta}\cdot r\,d\theta\,\hat{\theta} = H(r)\,r\int_0^{2\pi} d\theta = H(r)\cdot 2\pi r$$

L.D.

$$I_{\substack{\text{enlazada}\\ \text{libre}}} = \int_0^{2\pi}\int_0^{a} \|J_1\|\hat{z}\cdot r\,dr\,d\theta\,\hat{z} + \int_0^{2\pi}\int_a^{b} \|J_2\|\hat{z}\cdot r\,dr\,d\theta\,\hat{z}$$

$$= \pi\|J_1\|a^2 + \|J_2\|\cdot\cancel{2}\pi\cdot\frac{1}{\cancel{2}}(b^2-a^2)$$

$$= \pi\|J_1\|a^2 + \|J_2\|\pi(b^2-a^2)$$

— o —

$$\Rightarrow H(r)\cdot 2\pi r = \pi\|J_1\|a^2 + \|J_2\|\pi(b^2-a^2)$$

$$\Rightarrow \boxed{\vec{H} = \frac{\|J_1\|a^2 + \|J_2\|(b^2-a^2)}{2}\,\frac{1}{r}\,\hat{\theta}} \quad (r>b) \qquad \boxed{\vec{M} = \chi\,\frac{\|J_1\|a^2 + \|J_2\|(b^2-a^2)}{2}\,\frac{1}{r}\,\hat{\theta}}$$

P3 c) $\vec{J}_m = \nabla \times \vec{M}$

$$\nabla \times \vec{M} = \frac{1}{r} \begin{array}{ccc} + & - & + \\ \end{array}$$
$$\nabla \times \vec{M} = \frac{1}{r} \begin{vmatrix} \hat{r} & r\hat{\theta} & \hat{z} \\ \partial_r & \partial_\theta & \partial_z \\ \cancelto{0}{M_r} & rM_\theta & \cancelto{0}{M_z} \end{vmatrix}$$

$$\boxed{\vec{M} = \frac{1}{2}\chi \|J_1\| r \hat{\theta}}, \quad 0<r<a$$

$$\nabla \times \vec{M} = \frac{1}{r}\left(0\hat{r} - 0\hat{\theta} + \left(\frac{\partial}{\partial r}\left(\frac{1}{2}\chi \|J_1\| r^2\right) - 0 \right)\hat{z} \right)$$

$$= \frac{1}{r}\,\frac{1}{2}\chi \|J_1\| \cdot 2r\,\hat{z}$$

$$= \boxed{\frac{1}{2}\chi \|J_1\| \hat{z} = \vec{J}_m}$$

$$\nabla\times\vec{M} = \frac{1}{r}\begin{array}{c} \begin{array}{ccc} + & \quad - & \quad + \end{array} \\ \begin{vmatrix} \hat{r} & r\hat{\theta} & \hat{z} \\ \partial_r & \partial_\theta & \partial_z \\ \cancelto{0}{M_r} & rM_\theta & \cancelto{0}{M_z} \end{vmatrix} \end{array}$$

$(a<r<b)$ $\boxed{\vec{M} = \chi\left(\frac{\|J_1\|a^2}{2}\,\frac{1}{r} + \frac{\|J_2\|}{2}\,\frac{r^2-a^2}{r}\right)\hat{\theta}}$

$$\nabla\times\vec{M} = \frac{1}{r}\left(0\hat{r} - 0\hat{\theta} + \frac{\partial}{\partial r}\left(\chi\frac{\|J_1\|a^2}{2} + \chi\frac{\|J_2\|}{2}(r^2-a^2) - 0\right)\hat{z}\right)$$

$$= \frac{1}{r}\,\chi\frac{\|J_2\|}{2}\cdot 2r\,\hat{z}$$

$$= \boxed{\chi\|J_2\|\hat{z} = \vec{J}_m}$$

$$\nabla \times \vec{M} = \frac{1}{r} \begin{array}{c} \begin{array}{ccc} + & - & + \end{array} \\ \begin{vmatrix} \hat{r} & r\hat{\theta} & \hat{z} \\ \partial_r & \partial_\theta & \partial_z \\ \cancelto{0}{M_r} & rM_\theta & \cancelto{0}{M_z} \end{vmatrix} \end{array}$$

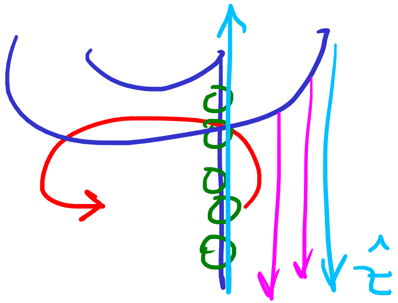
$(r>b)$

$$\boxed{\vec{M} = \chi \frac{\|J_1\| a^2 + \|J_2\|(b^2 - a^2)}{2} \frac{1}{r} \hat{\theta}}$$

$$\nabla \times \vec{M} = \frac{1}{r}\left(0\hat{r} - 0\hat{\theta} + (0-0)\hat{z}\right) = \boxed{\vec{0} = \vec{J}_m}$$

P3 d)

$$\vec{K}_m = \vec{M}\Big|_{sup} \times \hat{n}_{ext}$$



1. $\vec{K}_m = \vec{M}(r=a) \times \hat{r}$

2. $\vec{K}_m = \vec{M}(r=a) \times (-\hat{r})$

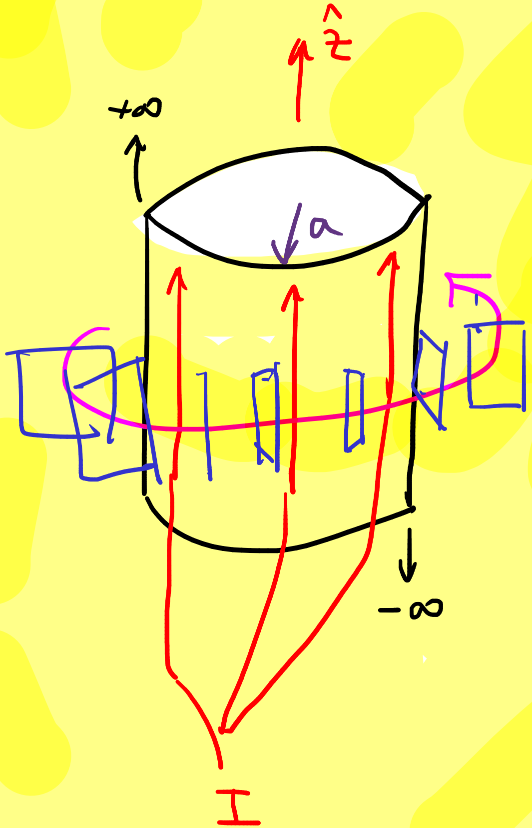
   $\vec{K}_m = \vec{M}(r=b) \times \hat{r}$

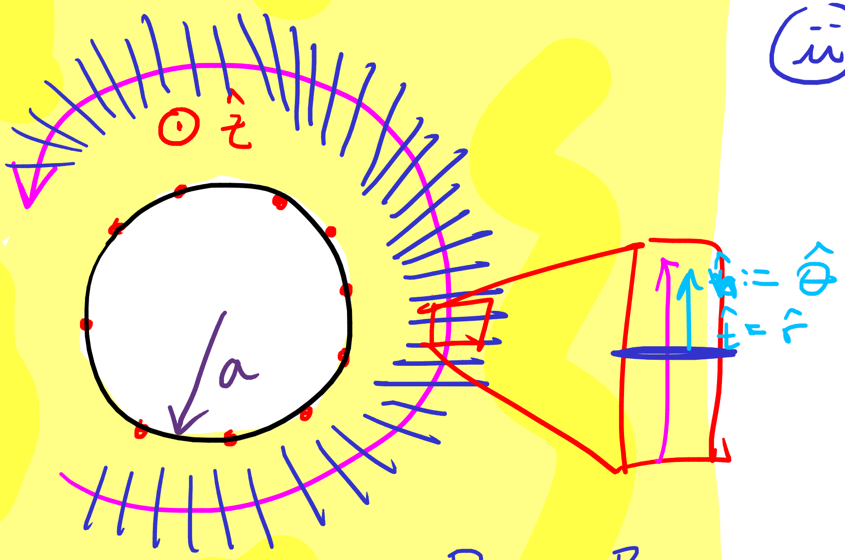
3. $\vec{K}_m = \vec{M}(r=b) \times (-\hat{r})$

P_4 AUX 17

P4. [Y si las cosas cambian]

Se tiene un conductor cilíndrico lo suficientemente largo por el cual circula una corriente I en su superficie en el sentido del eje. Obtenga expresiones para campo magnético, intensidad magnética, magnetización y corrientes de magnetización, si su exterior se sumerge en un medio de permeabilidad μ_1 que satisface la identidad $\frac{\mu_0}{\mu_1} = \beta\left(\text{sen}^2(\theta) + r\cos(\theta)\right)$. Justifique todos sus pasos.





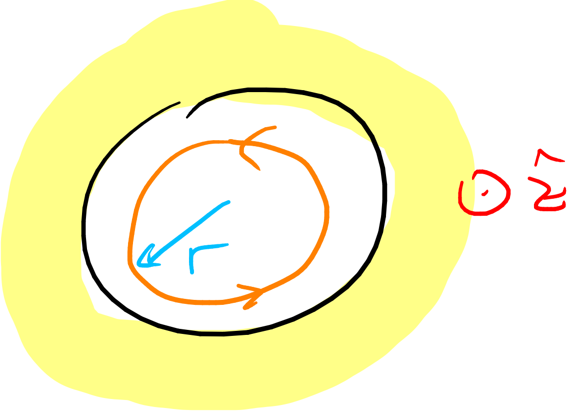
(i) $0 < r < a$

(ii) $r > a$

$$B_{1n} = B_{2n} \Rightarrow B_{1\theta} = B_{2\theta} = B$$

(i) $0<r<a$

Ley de Ampère: $\oint_\Gamma \vec{B}\cdot d\vec{l} = \mu_0 I_{enlazada}$



L.I.

$$\oint_0^{2\pi} B(r)\hat{\theta}\cdot r\,d\theta\,\hat{\theta} = B(r)\cdot 2\pi r$$

L.D.

$I_{enlazada} = 0$

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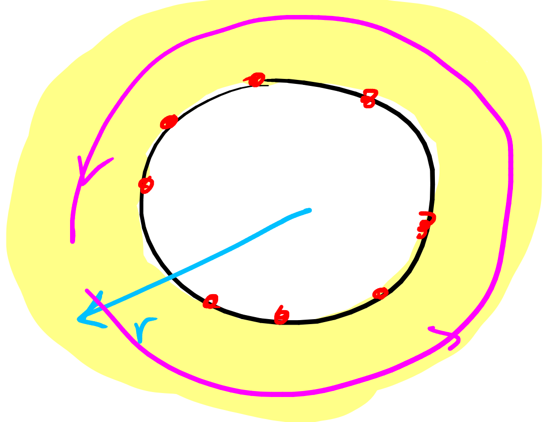
$\Rightarrow \vec{B} = \vec{0}$ $\qquad \mu\vec{H} = \vec{B}$

$\Rightarrow \vec{H} = \vec{0}$ $\qquad \vec{M} = \chi\vec{H} = \left(\frac{\mu}{\mu_0} - 1\right)\vec{H}$

$\Rightarrow \vec{M} = \vec{0}$

$\vec{J}_M = \vec{0} = \vec{K}_M$

ii) $\boxed{r > a}$



Ley de Ampère: $\oint_\Gamma \vec{B} \cdot d\vec{l} = \mu_0 \underbrace{I_{enlazada}}_{I_{libre} + I_m}$

Ley de Ampère generalizada: $\oint_\Gamma \vec{H} \cdot d\vec{l} = I_{enlazada\ libre}$

$\vec{B} = B(r,\theta)\hat{\theta}$ pues $\vec{B} \propto \mu_0$ y $\mu_0 = \mu_0(r,\theta)$

$\frac{\mu_0}{\mu_1} = \beta\left(\operatorname{sen}^2(\theta) + r\cos(\theta)\right)$

y $\vec{H} = H(r,\theta)\hat{\theta}$ pues $\vec{H} = \frac{1}{\mu_1}\vec{B}$, $\mu_1 = \mu_1(r,\theta)$

$\mu_0 = \mu_0(r,\theta)$; $\mu_1 = \mu_1(r,\theta)$

L.I.

$$\oint_\Gamma \vec{H}\cdot d\vec{l} = \int_0^{2\pi} H(r,\theta)\hat{\theta}\cdot r\,d\theta\,\hat{\theta} = \int_0^{2\pi} \frac{1}{\mu_1} B(r,\theta)\, r\,d\theta = \int_0^{2\pi} \frac{1}{\mu_0}\beta\left(\text{sen}^2(\theta) + r\cos(\theta)\right) B(r,\theta)\, r\,d\theta$$

$$\vec{H} = \frac{1}{\mu_1}\vec{B}$$

$$\int_0^{2\pi} \text{sen}^2(\theta)\,d\theta = \int_0^{2\pi} \frac{1-\cos(2\theta)}{2}\,d\theta = \frac{1}{2}\int_0^{2\pi} d\theta - \frac{1}{2}\int_0^{2\pi}\cos(2\theta)\,d\theta$$

$$= \frac{1}{2}\left[\theta\right]_0^{2\pi} - \frac{1}{2}\cdot\frac{1}{2}\left[\text{sen}(2\theta)\right]_0^{2\pi} = \pi$$

$$\int_0^{2\pi}\cos(\theta)\,d\theta = \left[\text{sen}(\theta)\right]_0^{2\pi} = 0$$

$$= \int_0^{2\pi} \frac{1}{\mu_0}\beta\left(\text{sen}^2(\theta) + r\cos(\theta)\right) B(r)\, r\,d\theta$$

$$= \frac{1}{\mu_0}\beta B(r)\, r\left(\underbrace{\int_0^{2\pi}\text{sen}^2(\theta)\,d\theta}_{\pi} + r\underbrace{\int_0^{2\pi}\cos(\theta)\,d\theta}_{0}\right)$$

$$= \beta\pi\frac{1}{\mu_0}B(r)\,r$$

L.D.

$$I_{\text{enlazada libre}} = I$$

—o—

$$\Rightarrow \beta\pi\frac{1}{\mu_0}B(r)\,r = I \Rightarrow \boxed{\vec{B}(r) = \frac{I\mu_0}{\beta\pi}\frac{1}{r}\,\hat{\theta}}, \quad r>a$$

$$\boxed{\vec{H} = \frac{1}{\cancel{\mu_0}} \beta\left(\operatorname{sen}^2(\theta) + r\cos(\theta)\right) \cdot \frac{I\cancel{\mu_0}}{\beta\pi}\frac{1}{r}\hat{\theta}} \qquad r > a$$

$$\vec{M} = \left(\frac{\mu_1}{\mu_0} - 1\right)\vec{H} = \left(\frac{\mu_1}{\mu_0} - 1\right)\frac{1}{\mu_1}\vec{B}$$

$$= \left(\frac{1}{\mu_0} - \frac{1}{\mu_1}\right)\vec{B}$$

$$= \left(\frac{1}{\mu_0} - \frac{1}{\mu_0}\beta\left(\operatorname{sen}^2(\theta) + r\cos(\theta)\right)\right) \cdot \frac{I\mu_0}{\beta\pi}\frac{1}{r}\hat{\theta}$$

$$\boxed{\vec{M} = \left(1 - \beta\left(\operatorname{sen}^2(\theta) + r\cos(\theta)\right)\right) \cdot \frac{I}{\beta\pi}\frac{1}{r}\hat{\theta}} \qquad r > a$$

$$\vec{J}_m = \nabla \times \vec{M}$$

$$\nabla \times \vec{M} = \frac{1}{r} \begin{array}{ccc} + & - & + \\ \end{array}$$

$$\nabla \times \vec{M} = \frac{1}{r} \begin{vmatrix} \hat{r} & r\hat{\theta} & \hat{z} \\ \partial_r & \partial_\theta & \partial_z \\ \cancelto{0}{M_r} & rM_\theta & \cancelto{0}{M_z} \end{vmatrix}$$

$$= \frac{1}{r}\left(0\hat{r} - 0\hat{\theta} + \left(\frac{\partial}{\partial r}\left(rM_\theta\right) - 0\right)\hat{z}\right)$$

$$= \frac{1}{r}\frac{\partial}{\partial r}\left(\left(1-\beta\left(\text{sen}^2(\theta)+r\cos(\theta)\right)\right)\cdot\frac{I}{\beta\pi}\right)\hat{z}$$

$$= \boxed{\frac{1}{r}\cdot\frac{I}{\beta\pi}\cdot\cos(\theta)\,\hat{z} = \vec{J}_m}$$

$$\vec{K}_m = \vec{M}\big|_{sup} \times \hat{n}_{ext}$$

$$= \vec{M}(r=a) \times (-\hat{r})$$

$$= \left(1-\beta\left(sen^2(\theta)+a\cos(\theta)\right)\right) \cdot \frac{I}{\beta\pi} \frac{1}{a} \hat{\theta} \times (-\hat{r})$$

$$\boxed{= \left(\beta\left(sen^2(\theta)+a\cos(\theta)\right)-1\right) \frac{I}{\beta\pi} \hat{z} = \vec{K}_m}$$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